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Systems and methods for data transfer for computational storage devices

Abstract

Provided are systems, methods, and apparatuses for managing memory. The method can include: establishing a connection via an interface, between a host device and a storage device; and transferring data, via the interface, between first memory associated with the host device and second memory associated with the storage device by performing a data operation on the second memory by an application executed by the host, where the storage device includes a processing element that accelerates the data operation by performing at least one offload function on the data operation.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5805777	12/1997	Kuchta	N/A	N/A
6976255	12/2004	Clark et al.	N/A	N/A
7315807	12/2007	Lavallee et al.	N/A	N/A
7433931	12/2007	Richoux	N/A	N/A
7801931	12/2009	Tunar et al.	N/A	N/A
7937703	12/2010	Bethea et al.	N/A	N/A
8381004	12/2012	Elnozahy et al.	N/A	N/A
8615570	12/2012	Brewis et al.	N/A	N/A
8726299	12/2013	Bromley et al.	N/A	N/A
8904072	12/2013	Lee	N/A	N/A
8943504	12/2014	Vijayarajan et al.	N/A	N/A
9203627	12/2014	Edstrom et al.	N/A	N/A

9398016	12/2015	Chakraborty et al.	N/A	N/A
9406082	12/2015	Smedberg et al.	N/A	N/A
9596135	12/2016	Thomas et al.	N/A	N/A
9626120	12/2016	Jia et al.	N/A	N/A
10162793	12/2017	Bshara et al.	N/A	N/A
10303382	12/2018	Chu et al.	N/A	N/A
10359953	12/2018	Cargnini	N/A	G06F 3/065
10394746	12/2018	Kachare et al.	N/A	N/A
10404811	12/2018	Ly et al.	N/A	N/A
10412187	12/2018	Busayarat et al.	N/A	N/A
10476936	12/2018	Retnakumari et al.	N/A	N/A
10521114	12/2018	Thomas	N/A	N/A
10621114	12/2019	Bshara et al.	N/A	N/A
10637962	12/2019	Gay et al.	N/A	N/A
10771340	12/2019	Ballapuram	N/A	N/A
10795726	12/2019	Fujii et al.	N/A	N/A
10802753	12/2019	Kabra et al.	N/A	N/A
10838905	12/2019	Zou et al.	N/A	N/A
10846096	12/2019	Chung et al.	N/A	N/A
10860511	12/2019	Thompson	N/A	G06F 13/28
10877750	12/2019	Connolly et al.	N/A	N/A
11294576	12/2021	Khan	N/A	G06F 3/0643
11334382	12/2021	Bernat et al.	N/A	N/A
11474974	12/2021	Lutz et al.	N/A	N/A
11507298	12/2021	Yang	N/A	G06F 3/0679
11630921	12/2022	Sinha et al.	N/A	N/A
11687498	12/2022	Trika et al.	N/A	N/A
2003/0093572	12/2002	Laux et al.	N/A	N/A
2004/0220918	12/2003	Scriffignano et al.	N/A	N/A
2005/0132341	12/2004	Lanzatella et al.	N/A	N/A
2007/0276924	12/2006	Newton et al.	N/A	N/A
2008/0004887	12/2007	Brunswick et al.	N/A	N/A
2008/0279161	12/2007	Stirbu et al.	N/A	N/A
2011/0066676	12/2010	Kleyzit et al.	N/A	N/A
2011/0185295	12/2010	Kashida	N/A	N/A
2012/0079172	12/2011	Yoshida	N/A	N/A
2012/0152576	12/2011	Bassinger et al.	N/A	N/A
2012/0162234	12/2011	Blinzer	345/501	G06T 1/20
2013/0173662	12/2012	Kaplinger et al.	N/A	N/A
2014/0143778	12/2013	Gopal et al.	N/A	N/A
2014/0310332	12/2013	Huang et al.	N/A	N/A
2015/0319246	12/2014	Ishizaka	709/217	H04L 67/1097
2016/0012128	12/2015	Siebler et al.	N/A	N/A
2016/0054931	12/2015	Romanovsky	711/103	G06F 3/0643
2016/0328344	12/2015	Jose et al.	N/A	N/A
2017/0249088	12/2016	Thomas	N/A	N/A
2018/0063145	12/2017	Cayton et al.	N/A	N/A
2018/0352038	12/2017	Sathyanarayana et al.	N/A	N/A
2019/0037280	12/2018	Yang et al.	N/A	N/A
2019/0349281	12/2018	Oztaskent et al.	N/A	N/A
2020/0012439	12/2019	Khan et al.	N/A	N/A
2020/0050480	12/2019	Cao	N/A	G06F 3/067
2020/0057578	12/2019	Benisty et al.	N/A	N/A
2020/0081640	12/2019	Enz et al.	N/A	N/A
2020/0117517	12/2019	Costa	N/A	G06F 9/546
2020/0167098	12/2019	Shah et al.	N/A	N/A
2020/0183582	12/2019	Kachare et al.	N/A	N/A
2020/0192592	12/2019	Song et al.	N/A	N/A
2020/0201575	12/2019	Mizrahi	N/A	N/A
2020/0301898	12/2019	Samynathan	N/A	G06F 16/2453
2020/0310694	12/2019	Gao	N/A	G06F 3/0683
2020/0379781	12/2019	Rachapudi et al.	N/A	N/A
2021/0026560	12/2020	Pillai	N/A	G11C 16/20
2021/0247922	12/2020	Song et al.	N/A	N/A
2021/0303159	12/2020	Ke	N/A	G06F 12/0246
2021/0377117	12/2020	Kanevsky	N/A	H04L 41/0843
2022/0058137	12/2021	Xu	N/A	G06F 13/4221
2022/0188028	12/2021	Mesnier	N/A	G06F 3/061
2022/0236911	12/2021	Jones	N/A	G06F 3/0679
2022/0295255	12/2021	Karampatsis et al.	N/A	N/A
2023/0244499	12/2022	Gupta et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102929813	12/2015	CN	N/A

109254873	12/2018	CN	N/A
3964956	12/2021	EP	N/A
20120071060	12/2011	KR	N/A
20140001970	12/2013	KR	N/A
20170121661	12/2016	KR	N/A
20180106822	12/2017	KR	N/A
20200068564	12/2019	KR	N/A
200622673	12/2005	TW	N/A
201214288	12/2011	TW	N/A
201324352	12/2012	TW	N/A
202102976	12/2020	TW	N/A
2002097610	12/2001	WO	N/A

OTHER PUBLICATIONS

Office Action for U.S. Appl. No. 17/234,780, mailed Jan. 5, 2023. cited by applicant

Anandan, Sabby et al., "Industry Paper: Spring XD—A Modular Distributed Stream and Batch Processing System," DEBS '15: Proceedings of the 9th ACM I/O Workshop on Data Management in End-Scale Systems, Jun. 2015, pp. 217-225. cited by applicant

Carbone, Paris et al., "Apache Flink™: Stream and Batch Processing in a Single Engine," Bulletin of the IEEE Computer Society Technical Committee on Data Engineering, vol. 34, No. 1, Jan. 2014, pp. 34-43. cited by applicant

European Extended Search Report for Application No. 21190626.8, mailed Jan. 28, 2022. cited by applicant

European Extended Search Report for Application No. 21212137.0, mailed Apr. 28, 2022. cited by applicant

Fisher, Mark et al., "Spring XD Guide", 1.3.3.BUILD-SNAPSHOT, 2013, 400 pages. cited by applicant

NVM Express, Inc., "NVM Express TM over Fabrics Revision 1.1," Oct. 22, 2019, pp. 1-83, XP055877791, Retrieved from the Internet: URL:https://nvmexpress.org/specifications/NVM-Express-1.1-2019.10.22-Ratified.pdf, retrieved on Jan. 11, 2022. cited by applicant

Torabzadehkashi, Mahdi, et al. "Computational Storage: An Efficient and Scalable Platform for Big Data and HPC Applications," Journal of Big Data, vol. 6, No. 1, 2019, pp. 1-15. cited by applicant

Final Office Action for U.S. Appl. No. 17/234,780, mailed Aug. 17, 2023. cited by applicant

Office Action for U.S. Appl. No. 17/234,780, mailed Dec. 22, 2023. cited by applicant

Office Action for U.S. Appl. No. 17/359,495, mailed Dec. 7, 2023. cited by applicant

Office Action for U.S. Appl. No. 17/502,023, mailed Oct. 5, 2023. cited by applicant

Bates, Stephen, "Accelerating RocksDB with Eideticom's NoLoad NVMe-based Computational Storage Processor," Storage Developer Conference 2019, Sep. 2019, retrieved from the Internet: URL:https://www.snia.org/sites/default/files/SDC/2019/presentations/Computational/Bates_SStephen_Accelerating_RocksDB_with_Eideticom's_NoLoad_NVMe-based_Computational_Storage_Processor.pdf [retrieved on Feb. 8, 2024], 16 pages. cited by applicant

Chambers, Bill et al., "Spark, The Definitive Guide: Big Data Processing Made Simple", first 400 pages, Feb. 2018, retrieved from the Internet: URL:https://analyticsdata24.files.wordpress.com/2020/02/spark-the-definitive-guide400www.bigdatabugs.com_.pdf [retrieved on Jan. 18, 2024], 400 pages. cited by applicant

European Summons to Oral Proceedings for Application No. 21190626.8, mailed Feb. 15, 2024. cited by applicant

European Summons to Oral Proceedings for Application No. 21190952.8, mailed Jan. 25, 2024. cited by applicant

Gounaris, Anastasios et al., "Dynamic Configuration of Partitioning in Spark Applications," IEEE Transactions on Parallel and Distributed Systems, vol. 28, No. 1, Jan. 2017, pp. 1-15. cited by applicant

Molgaard, Jason, "A Simple Approach to Implementing Computational Storage," Storage Developer Conference 2019, Sep. 2019, retrieved from the Internet: URL:https://www.snia.org/sites/default/files/SDC/2019/presentations/Computational/Molgaard_Jason_A_Simple_Approach_to_Implementing_Computational_Storage.pdf [retrieved on Jan. 11, 2022], 25 pages. cited by applicant

Prasaad, Guna et al., "Improving High Contention OLTP Performance via Transaction Scheduling", arXiv.org, Oct. 2018, retrieved from the Internet: URL:https://arxiv.org/abs/1810.08111 [retrieved on Jan. 17, 2024], 15 pages. cited by applicant

Sabiu, Habib Ado, "Performance Evaluation of Job Scheduling and Resource Allocation in Apache Spark," Jul. 2018, retrieved from the Internet: URL:https://harvest.usask.ca/server/api/core/bitstreams/9654d97a-5217-486d-b8ae-df305acfc5b8/content [retrieved on Jan. 17, 2024], 104 pages. cited by applicant

Shadley, Scott et al., "What Happens When Compute Meets Storage?," Sep. 2019, retrieved from the Internet: URL:https://www.snia.org/sites/default/files/SDC/2019/presentations/Computational/Shadley_Scott_Adams_Nick_What_Happens_when_Compute_Meets_Storage.pdf [retrieved on Jan. 13, 2022], 28 pages. cited by applicant

SNIAVIDEO: "What Happens when Compute Meets Storage (SDC 2019)," Oct. 2019, retrieved from the Internet: URL:https://www.youtube.com/watch?v=45v8v8v8v8 [retrieved on Jan. 13, 2022], 28 pages. cited by applicant

Tang, Shanjiang et al., "A Survey on Spark Ecosystem for Big Data Processing," arxiv.org, Nov. 2018, 21 pages. cited by applicant

Wikipedia, "Message Broker," Jul. 2020, retrieved from the Internet: URL:https://en.wikipedia.org/w/index.php?title=Message_broker&oldid=966520924 [retrieved on Jan. 13, 2022], 28 pages. cited by applicant

Final Office Action for U.S. Appl. No. 17/234,780, mailed Jul. 5, 2024. cited by applicant

Final Office Action for U.S. Appl. No. 17/502,023, mailed Jun. 20, 2024. cited by applicant

Office Action for U.S. Appl. No. 17/359,495, mailed Aug. 12, 2024. cited by applicant

Castellani, Angelo P. et al., "WebIoT: A Web Application Framework for the Internet of Things," WCNC 2012 Workshop on Internet of Things Enabling Technologies, Communications and Beyond, 2012, pp. 202-207. cited by applicant

European Office Action for Application No. 22174076.4, mailed Jul. 8, 2024. cited by applicant

Kerdoudi, Mohamed Lamine et al., "Recovering Software Architecture Product Lines," 2019 24th International Conference on Engineering of Complex Computer Systems, 2019, pp. 1-10. cited by applicant

European Office Action for Application No. 21190952.8, mailed Oct. 18, 2024. cited by applicant

European Office Action for Application No. 21212137.0, mailed Oct. 25, 2024. cited by applicant

Li, Yonghui et al., "Architecture and Analysis of a Dynamically-Scheduled Real-Time Memory Controller," Real-Time Systems, Jul. 2015, vol. 52, pp. 675-720. cited by applicant

Li, Yonghui, "Design and Formal Analysis of Real-Time Memory Controllers," Phd Thesis 1 (Research TU/e /Graduation TU/e), Electrical Engineering, Sep. 2014, 100 pages. cited by applicant

Mutlu, Onur et al., "Parallelism-Aware Batch Scheduling: Enhancing both Performance and Fairness of Shared DRAM Systems," International Symposium on Memory Management, 2008, pp. 63-74. cited by applicant

Prieto, Pablo et al., "CMP Off-Chip Bandwidth Scheduling Guided by Instruction Criticality," ICS '13: Proceedings of the 27th International ACM Conference on Supercomputing, Jun. 2013, pp. 379-388. cited by applicant

Zhou, Ping et al., "Throughput Enhancement for Phase Change Memories," IEEE Transactions on Computers, vol. 63, No. 8, Aug. 2014, pp. 2080-2093. cited by applicant

Notice of Allowance for U.S. Appl. No. 17/502,023, mailed Nov. 26, 2024. cited by applicant

Office Action for U.S. Appl. No. 17/234,780, mailed Dec. 19, 2024. cited by applicant

Warke, Amit, et al. "Storage Service Orchestration with container Elasticity," 2018 IEEE 4th International Conference on Collaboration and Internet Computing, 2018, pp. 1-6. cited by applicant

Notice of Allowability for U.S. Appl. No. 17/502,023, mailed Feb. 4, 2025. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/241,525, filed Sep. 7, 2021; U.S. Provisional Patent Application Ser. No. 63/144,469, filed Feb. 1, 2021; and U.S. Provisional Patent Application Ser. No. 63/142,485, filed Jan. 27, 2021, which are incorporated by reference herein for all purposes.

FIELD

(1) The disclosure relate generally to a storage system, and more particularly to systems and methods for data transfer for computation storage devices.

BACKGROUND

(2) The present background section is intended to provide context only, and the disclosure of any concept in this section does not constitute an admission that said concept is prior art.

(3) With advances in technology, the size and amount of data is increasing rapidly as data is collected by devices such as mobile devices, Internet of things devices, aerial (remote sensing) devices, software logs, cameras, microphones, radio-frequency identification (RFID) readers, wireless sensor networks, and the like. To process and use information represented by the collected data, storage devices, processing elements, and servers are often used in datacenters to filter, compute, store, and perform related operations on the data. A need remains for systems and methods that improve data transfers in storage devices.

(4) The above information disclosed in this Background section is only for enhancement of understanding of the background of the disclosure and therefore it may contain information that does not constitute prior art.

SUMMARY

(5) In various embodiments, described herein include systems, methods, and apparatuses for data transfers for computation storage devices.

(6) In various embodiments, a method to manage memory is described. The method includes: establishing a connection, via an interface, between a host device and a storage device; and transferring data, via the interface, between first memory associated with the host device and second memory associated with the storage device by performing a data operation on the second memory by an application executed by the host, where the storage device includes a processing element that accelerates the data operation by performing at least one offload function on the data operation.

(7) In one embodiment, the interface includes an application programming interface (API), and the host device allocates the second memory using at least one command associated with the API. In another embodiment, the host device allocates the second memory using a fine-grained allocation scheme or a coarse-grained allocation scheme. In one embodiment, the method further includes addressing the second memory and the transferring data is based at least in part on the addressing of the second memory. In some embodiments, second memory includes a secure address space, and where the method further includes host device performing one or more multitenant operations based on the secure address space.

(8) In another embodiment, the storage device includes a non-volatile memory express (NVMe) enabled storage device and the data operation is performed using a scatter gather list (SGL). In some embodiments, the performing the data operation further includes the storage device transferring data from persistent memory associated with the storage device to the second memory. In one embodiment, the transferring data further includes the host device providing information regarding host device layout to the storage device. In another embodiment, the transferring data further includes the storage device providing information regarding the storage device's memory layout for read and write operations to the host device. In some embodiments, the establishing the connection includes a discovery and setup process.

(9) Similarly, non-transitory computer-readable media, devices, and systems for performing substantially the same or similar operations as described above are further disclosed.

(10) Accordingly, particular embodiments of the subject matter described herein can be implemented so as to realize one or more of the following advantages. Reduce network latencies and improve network stability and operational data transfer rates and, in turn, improve the user experience. Reduce costs associated with routing network traffic, network maintenance, network upgrades, and/or the like. Further, in some aspects, the disclosed systems can serve to reduce the power consumption and/or bandwidth of devices on a network, and may serve to increase the speed and/or efficiency of communications between devices. Moreover, the disclosed systems and methods can serve to provide a seamless mechanism that works for direct-attached and network-attached drives. In another aspect, the disclosed mechanisms can provide scaling at least because the device memory may not be exposed to host. Further, the disclosed mechanisms can be applied to a computational storage device (e.g., a storage device having a processing functionality) and related systems. As noted, the disclosed systems have the advantage of not exposing device memory. Thus, the disclosed systems do not necessarily need to have their memory be populated by system software in host's system virtual address space. In other aspects, the disclosed systems can be used in connection with any suitable storage device such as a non-NVMe solid state drives (SSDs). Further, the disclosed systems and methods can be standardized and applied to a range of storage devices and related systems while being agnostic to vendor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above-mentioned aspects and other aspects of the present techniques will be better understood when the present application is read in view of the following figures in which like numbers indicate similar or identical elements. Further, the drawings provided herein are for purpose of illustrating certain embodiments only; other embodiments, which may not be explicitly illustrated, are not excluded from the scope of this disclosure.

(2) These and other features and advantages of the present disclosure will be appreciated and understood with reference to the specification, claims, and appended drawings wherein:

(3) FIG. 1 is an illustration of an exemplary architecture for enabling data processing functions to be offloaded to a storage device, in accordance with example embodiments of the disclosure.

(4) FIG. 2 is an illustration of an exemplary schematic of memory allocations with respective offsets in a device memory space, in accordance with example embodiments of the disclosure.

- (5) FIG. 3 is an illustration of an exemplary schematic describing aspects of device managed device memory, in accordance with example embodiments of the disclosure.
- (6) FIG. 4 is an illustration of an exemplary data structure device may advertise its memory management and usage capabilities, in accordance with example embodiments of the disclosure.
- (7) FIG. 5 is an illustration of an exemplary commands for using device memory address for direct memory access (DMA), in accordance with example embodiments of the disclosure.
- (8) FIG. 6 is an illustration of an exemplary non-volatile memory express (NVMe) command contains a scatter gather list (SGL) having corresponding fields, in accordance with example embodiments of the disclosure.
- (9) FIG. 7 is an illustration of an exemplary type of SGL mechanism that is employed to keep memory usage similar for local and remote-attached computational storage devices, in accordance with example embodiments of the disclosure.
- (10) FIG. 8 is an illustration of an exemplary table that includes the interaction of data buffers with I/O commands, in accordance with example embodiments of the disclosure.
- (11) FIG. 9 is an illustration of an exemplary SGL descriptor for data transfer, in accordance with example embodiments of the disclosure.
- (12) FIG. 10 is an illustration of an exemplary operational flow illustrating example operations associated with the disclosed systems, in accordance with example embodiments of the disclosure.
- (13) FIG. 11 is an illustration of another exemplary operational flow illustrating example operations associated with the disclosed systems, in accordance with example embodiments of the disclosure.
- (14) FIG. 12 shows an example schematic diagram of a system that can be used to practice embodiments of the present disclosure.
- (15) FIG. 13 shows an example schematic diagram of a management computing entity, in accordance with example embodiments of the disclosure.
- (16) FIG. 14 shows an example schematic diagram of a user device, in accordance with example embodiments of the disclosure.
- (17) While the present techniques are susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described. The drawings may not be to scale. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the present techniques to the particular form disclosed, but to the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present techniques as defined by the appended claims.

DETAILED DESCRIPTION OF VARIOUS EMBODIMENTS

- (18) The details of one or more embodiments of the subject matter described herein are set forth in the accompanying drawings and the description below. Other features, aspects, and advantages of the subject matter will become apparent from the description, the drawings, and the claims.
- (19) Various embodiments of the present disclosure now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all embodiments are shown. Indeed, the disclosure may be embodied in many different forms and should not be construed as limited to the embodiments set forth herein; rather, these embodiments are provided so that this disclosure will satisfy applicable legal requirements. The term “or” is used herein in both the alternative and conjunctive sense, unless otherwise indicated. The terms “illustrative” and “example” are used to be examples with no indication of quality level. Like numbers refer to like elements throughout. Arrows in each of the figures depict bi-directional data flow and/or bi-directional data flow capabilities. The terms “path,” “pathway” and “route” are used interchangeably herein.
- (20) Embodiments of the present disclosure may be implemented in various ways, including as computer program products that comprise articles of manufacture. A computer program product may include a non-transitory computer-readable storage medium storing applications, programs, program components, scripts, source code, program code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like (also referred to herein as executable instructions, instructions for execution, computer program products, program code, and/or similar terms used herein interchangeably). Such non-transitory computer-readable storage media include all computer-readable media (including volatile and non-volatile media).
- (21) In one embodiment, a non-volatile computer-readable storage medium may include a floppy disk, flexible disk, hard disk, solid-state storage (SSS) (for example a solid-state drive (SSD)), solid state card (SSC), solid state module (SSM), enterprise flash drive, magnetic tape, or any other non-transitory magnetic medium, and/or the like. A non-volatile computer-readable storage medium may also include a punch card, paper tape, optical mark sheet (or any other physical medium with patterns of holes or other optically recognizable indicia), compact disc read only memory (CD-ROM), compact disc-rewritable (CD-RW), digital versatile disc (DVD), Blu-ray disc (BD), any other non-transitory optical medium, and/or the like. Such a non-volatile computer-readable storage medium may also include read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory (for example Serial, NAND, NOR, and/or the like), multimedia memory cards (MMC), secure digital (SD) memory cards, SmartMedia cards, CompactFlash (CF) cards, Memory Sticks, and/or the like. Further, a non-volatile computer-readable storage medium may also include conductive-bridging random access memory (CBRAM), phase-change random access memory (PRAM), ferroelectric random-access memory (FeRAM), non-volatile random-access memory (NVRAM), magnetoresistive random-access memory (MRAM), resistive random-access memory (RRAM), Silicon-Oxide-Nitride-Oxide-Silicon memory (SONOS), floating junction gate random access memory (FJG RAM), Millipede memory, racetrack memory, and/or the like.
- (22) In one embodiment, a volatile computer-readable storage medium may include random access memory (RAM), dynamic random access memory (DRAM), static random access memory (SRAM), fast page mode dynamic random access memory (FPM DRAM), extended data-out dynamic random access memory (EDO DRAM), synchronous dynamic random access memory (SDRAM), double data rate synchronous dynamic random access memory (DDR SDRAM), double data rate type two synchronous dynamic random access memory (DDR2 SDRAM), double data rate type three synchronous dynamic random access memory (DDR3 SDRAM), Rambus dynamic random access memory (RDRAM), Twin Transistor RAM (TTRAM), Thyristor RAM (T-RAM), Zero-capacitor (Z-RAM), Rambus in-line memory component (RIMM), dual in-line memory component (DIMM), single in-line memory component (SIMM), video random access memory (VRAM), cache memory (including various levels), flash memory, register memory, and/or the like. It will be appreciated that where embodiments are described to use a computer-readable storage medium, other types of computer-readable storage media may be substituted for or used in addition to the computer-readable storage media described above.
- (23) As should be appreciated, various embodiments of the present disclosure may also be implemented as methods, apparatus, systems, computing devices, computing entities, and/or the like. As such, embodiments of the present disclosure may take the form of an apparatus, system, computing device, computing entity, and/or the like executing instructions stored on a computer-readable storage medium to perform certain steps or operations. Thus, embodiments of the present disclosure may also take the form of an entirely hardware embodiment, an entirely computer program product embodiment, and/or an embodiment that comprises combination of computer program products and hardware performing certain steps or operations.
- (24) Embodiments of the present disclosure are described below with reference to block diagrams and flowchart illustrations. Thus, it should be understood that each block of the block diagrams and flowchart illustrations may be implemented in the form of a computer program product, an entirely hardware embodiment, a combination of hardware and computer program products, and/or apparatus, systems, computing devices, computing entities, and/or the like carrying out instructions, operations, steps, and similar words used interchangeably (for example the executable instructions, instructions for execution, program code, and/or the like) on a computer-readable storage medium for execution. For example, retrieval, loading, and execution of code may be performed sequentially such that one instruction is retrieved, loaded, and executed at a time. In some example embodiments, retrieval, loading, and/or execution may be performed in parallel such that multiple instructions are retrieved, loaded, and/or executed together. Thus, such embodiments can produce specifically-configured machines performing the steps or operations specified in the block diagrams

and flowchart illustrations. Accordingly, the block diagrams and flowchart illustrations support various combinations of embodiments for performing the specified instructions, operations, or steps.

Background and Problem Statement

(25) Computational storage devices can serve to enhance storage functionality uses, for example, by performing near-storage acceleration operations. In particular, the disclosed systems can perform accelerator device-based processing of storage data. In some respects, storage devices such as solid state storage devices (SSDs) can be configured to operate with any suitable protocol such as non-volatile memory express (NVMe). Such NVMe-enabled SSDs may have lower latencies than non-NVMe SSDs and can support greater storage capacities. In other respects, such SSDs can include multiple device input-output (I/O) queues, and thereby provide greater parallelism for storage. In some aspects, some Peripheral Component Interconnect Express (PCIe) based devices (e.g., PCIe generation 3 (Gen3) having 4 lanes or 8 lanes (x4 and x8 devices)) may saturate corresponding PCIe bandwidth limits and may thus limit device scaling. Accordingly, even if the SSD can provide a certain level of performance, the SSD may not be able to be used when multiple such devices are utilized in a server due to various system design limitations. Further, scaling SSDs in a server in connection with some PCIe generations (for example generation 4, Gen4) may have similar behaviors, for example, since the SSD may have higher performance and may saturate the PCIe limits per device. This similar behavior may be due to the PCIe lane count across devices; however, the SSDs may be limited by the total addressability of modern CPUs and the intended PCIe bus. (The above generations and corresponding lane counts are used as examples and merely for the purpose of illustration, and are not to be construed as limiting to the disclosed embodiments in any way.)

(26) With computational storage device peer-to-peer (P2P) technology, a processing element (e.g., a field-programmable gate array, FPGA) device memory can be mapped to a host address space; thus, an application can use this mapped address and read or write data directly to and from the SSD bypassing the host PCIe bus. Such a technique may work in situations in which the corresponding server supports such mapping. However, some servers may not support such mapping. Accordingly, as the device memory size increases, servers set restrictions in their BIOS to not map such large memories to host address space. Additionally, as multiple devices are scaled, the server may stop performing additional mappings thus limiting the device's capability, and causing further scaling issues. This can be due to the fact that the memory address space may be preconfigured by the server vendor, thereby restricting scaling for devices with large memories. In some cases, the usage of devices with large memories may be limited to a few devices (e.g., 1-6 devices) per server. In various aspects, computational storage provides the means to expand that capability by P2P.

(27) In some aspects, with certain protocols such as NVMe over fabrics (NVMe-oF) protocols and devices, the SSD can support network-attached modes of operation (as opposed to device attached modes of operation) without exposing the underlying interface details. However, the computational storage device or any device that utilizes such a mechanism may not function adequately since the device's memory mapping may not apply on the network for remote devices.

Further Limitations

(28) In some aspects, device memory mapped to a host address space (e.g., as a PCIe base address register, BAR) can have various limitations, as further described below. In particular, one limitation may include the fact that not all basic input output systems (BIOSs) treat the extra device memory equally. This may be due to the fact that many BIOSs employ legacy code that typically deal with allocations that work well for the host's dynamic random access memory (DRAM). Further, some BIOSs may not tolerate large device memories and if detected disable that resource from being initialized. Accordingly, there may be commands (e.g., PCIe configuration commands) to set the memory size for such devices so that BIOSs that support such commands will work with them. Moreover, BIOSs may not provide enough memory head-room for device memories. Accordingly, when multiple device memories get initialized with multiple such devices, the multiple device memories can run into memory address limitations due to this issue. Device scaling may get limited to a smaller number than what the server can accommodate without these visible additional device memories. In various respects, some memory behind PCIe BARs may not work when these devices are remote fabric attached through a network such as Ethernet, remote direct memory access (RDMA) over Converged Ethernet (RoCE), InfiniB and, Fibre Channel (FC) and so forth. This may be due to the fact that the memory map may be a local resource only that primarily works with local attached PCIe.

Solution

(29) In various aspects, computational storage (also referred to as near-storage computing) can enable certain data processing functions to be offloaded to the storage device. In some respects, the offloaded functions are also called as acceleration engines or acceleration kernels. Moreover, the offloaded functions can have compute resources and device local memory resources to perform the desired data processing functions. In another aspect, there can be multiple offloaded functions present and active in such a storage device. Such architectures may need a low overhead device memory management mechanism. The disclosure proposes multiple methods of efficient device memory management for these purposes.

(30) In various embodiments, the disclosed systems and methods include mechanisms for a storage device such as an SSD to transfer data such as read and write data from the storage medium to a co-device's memory. In another embodiment, the disclosed systems and methods include a device mechanism to translate device memory layout for reads and writes. In some embodiments, the disclosed systems describe a mechanism for the host to describe internal device layout to storage device without host system address space mapping and host translations. In some respects, the device layout can include a device mechanism to translate device memory for internal transfers between one or more devices situated internally within the same device. The translations are not exposed to the host and thereby kept private and secured. Further, the translations may have a requirement to be mapped into host address space. The host can include a virtual mapping mechanism that can include an offset between the start of this device memory (e.g., at 0h) to the end of this device memory, which can denote the length of this memory. In one embodiment, the disclosed systems and methods include a mechanism to discover and setup this feature, and further include a mechanism to apply host logic transparently for device that is direct attached or network attached, and can be implemented at least in part on a storage device such as a computational SSD.

(31) In some embodiments, the disclosed systems and methods include transferring data between storage device and one or more co-devices for peer-to-peer (P2P) data transfers of read and write data within the device and without going to external connections such as a PCIe bus. In one embodiment, the disclosed systems provide internal device memory mapping to the storage device for P2P transfers, and such transfers can occur without involving host's system address mapping of virtual and physical addresses provided by an operating system memory manager. In another embodiment, the disclosed systems permit abstracting internal mapping for host software command descriptions. In one embodiment, the disclosed systems provide a mechanism to transfer data between a device and its co-device and the ability of this functionality to scale to the maximum supported storage devices without causing system limits. In some respects, the disclosed systems provide mechanism for transmitting data between a storage device (SSD) and its co-device without host system address translations in the I/O path.

(32) FIG. 1 is an illustration of an exemplary architecture for enabling data processing functions to be offloaded to a storage device, in accordance with example embodiments of the disclosure. Diagram 100 shows a diagram including the components involved in this architecture.

(33) In various embodiments, diagram 100 shows host 102 that includes application 106, system memory 108, and drivers 110. In some embodiments, application 106 can include any suitable program being executed or partially executed on the host. In another embodiment, system memory 108 can include DRAM, Static Random Access Memory (SRAM), flash memory or any suitable volatile memory and can further include both logical and physical forms of memory. In some embodiments, the drivers 110 can include software or firmware that has specific instructions for the host software to communicate with the a corresponding connection at an external device similar, but not necessarily identical to, storage device 104, to be described below.

(34) In various embodiments, storage device 104 can include a solid state drive (SSD) or otherwise include flash memory. Further details of storage device 104 are provided below. In some embodiments, storage device 104 can include a device manager 112, which can further serve as a host interface, direct memory access (DMA) engine, and a platform manager. In some embodiments, the device manager 112 can interface with the host 102 through drivers 110. In some embodiments, storage device 104 can include flash 114 or any other suitable form of non-volatile memory. Further,

the flash **114** can be connected to the device manager **112**.

(35) In some embodiments, storage device **104** can include device memory **116**, which can include any suitable volatile memory and can further include both logical and physical forms of memory. The device memory **116** can include free memory blocks which can be utilized by the device manager **112** during function offload as described further herein. In some embodiments, storage device **104** various offload functions **118** (e.g., offload function **1**, offload function **2**, . . . , offload function **n**), as described further below, and which can be connected to one another (via a offload function control bus) and to the device memory **116** (via a device memory bus).

(36) Having described the components and connections between components in diagram **100**, we now turn to a discussion of the functional aspects of the device. As shown in diagram **100**, an offloaded function **118** can access to the device memory **116** for at least the following purposes: 1) input source data for processing, 2) intermediate results storage, 3) final results storage, and 4) other functions. In some embodiments, a device manager **112** can perform offload manager or acceleration management in the storage device **104** can fetch data from host **102** system memory **108** and deposit in the device memory **116** so that the offloaded function **118** can process it. Similarly, the offload function **118** can pick the results data from the device memory **116** and deposit that data into the system (e.g., host) memory **108** or persistent memory (e.g., non-volatile memory) such as flash **114** media. The device manager **112** can perform offload management and can provide the device memory addresses or pointers for source and results using a control bus. These addresses can be written by the device manager **112** to a set of registers for this purpose that are accessible to the offloaded functions **118** over the control bus. In another embodiment, the device manager **112** performing offload management in the storage device **104** can also provide device memory pointers to the offloaded functions for scratch pad or temporary purposes.

(37) In various aspects, the device memory **116** may need to have a relatively low overhead memory management that provides functions such as, 1) allocation, 2) de-allocation, 3) free space tracking, 4) garbage collection, 5) memory protection, and 6) others. In one embodiment, such device memory **116** can be managed by the host **102** in a transport agnostic manner. Such a transport agnostic manner can include the disclosed systems treating the device memory **116** as a virtual independent or separate memory address space from the host **102** system memory **108** address space. In another embodiment, such device memory **117** can be fully managed by the storage device **104** itself and the host **102** can refer to the device memory **116** in an abstract manner as needed. The disclosure describes these methods of device memory management further below.

(38) Host Managed Device Memory

(39) In this method, the device memory **116** can be managed by the host **102**. In some embodiments, the memory allocations, deallocations, and the like can be performed by the host system software (e.g., applications **106**) and conveyed to the storage device **104** through storage commands such as NVMe computational storage commands. The device memory **116** is not mapped into the host **102** system memory **108** address map; rather, it is treated as a separate virtual address space. Various storage commands such as NVMe commands provide the type of the address of source data, or result destination, and device memory address can be one of the types. In this approach, the device memory management can be performed by the host **102**. Further, the host can manage the device memory as a flat memory address space starting from an absolute address of 0h that spans the length of the memory. In some respects, the device may place restrictions on page granularity and alignment when the memory is addressed. The host can discover these restrictions as part of discovering device memory and the host can comply with the restrictions while using the device for proper device usage. The device manager **112** can relay the memory pointers received from the host **102** to the offloaded functions **118**. In one embodiment, it may be possible for the device manager **112** to perform offload platform management to add or subtract some base addresses from such host-provided device memory addresses before passing them to the offloaded functions **118**.

(40) Device Managed Device Memory

(41) In an embodiment, the device memory **116** can be abstracted from the host's **102** point of view and can therefore be managed by the storage device **104** itself. Accordingly, the host system software (e.g., applications **106**) may not specify the actual device memory addresses during the offloaded function invocation. In particular, the device can provide an opaque handle to memory allocated and the host addresses this memory by the provided handle and an address offset within that handle which can start at memory address 0h. In various embodiments, based on the length and type of the source data and result destination pointers provided by the host **102** system software, the device manager **112** performs offload platform management and can perform device memory allocation and reservations. The device manager **112** can include a DMA engine that can then fetch the source data, if necessary, and deposit into the allocated device memory buffers. At that point, the device manager **112** may provide those allocated device memory addresses to the respective offloaded function **118**, for example, using a control bus. Such memory allocation can be performed during every runtime invocation of the offloaded function. Similarly when the offloaded function **118** completes the data processing, the DMA engine may move the results to the destination addresses outside of device memory **116**, if necessary. The DMA engine can then de-allocate the device memory buffers used by the offloaded function **118** during that invocation. The device manager **112** may perform device memory garbage collection at any time to create contiguous blocks of free space.

(42) In some embodiments, an offloaded function **118** may have the requirement that a certain amount of temporary memory space (e.g., so-called scratch pad memory) be allocated for its processing during every invocation. Such fixed memory requirements can be fulfilled by the device manager **112** in the form of offload function registration. In particular, when the offloaded function is loaded into the storage device **104**, the offloaded function **118** may communicate such fixed memory requirements upfront to the device manager **112**, for example, using the control bus. Further, the device manager **112** can fulfil those requests either permanently or prior to invocation using any suitable protocol.

(43) In another embodiment, it may also be possible that an offloaded function **118** may make temporary memory allocation requests during runtime execution of the function. In particular, an offloaded function **118** may not be able to determine the temporary memory requirements upfront, prior to an invocation. Such dynamic device memory allocation is handled through any suitable control protocol. Further, such dynamic memory allocation requests may be conveyed by the offloaded function **118** over the control bus to the device manager **112**. Moreover, the device manager **112** can respond to those requests appropriately over the control bus.

(44) As described earlier, the results of data processing by the offloaded functions **118** reside in the device memory **116** at the time of completion. Those results can be transferred to the appropriate destination location in the system memory **108** or persistent memory (e.g., flash **114**) by the DMA engine of the device manager **112**. However, it may be possible that the destination address type may indicate that the results be kept in the device memory **116** itself. Such a mechanism can be used by the disclosed systems for chaining or nesting of the offloaded functions **118**. When results of one offloaded functions are fed to another offloaded function, the disclosed systems can maintain the results in the device memory **116** and provide the same device memory address as input data source to the next offloaded function. Further, such a method avoids data copying which can add more time latency. In the situation where the destination address type is a device memory, the associated device memory block is not de-allocated, which would be the case otherwise.

(45) In one embodiment, the device manager **112** can manage the device memory **116** as a virtual address space or physical address space. In case of virtual address space the device manager **112** may perform address translation of the device memory **116** accesses by the offloaded functions **118**. Using virtual address space method the device manager **112** may be able to offer a contiguous address space to the offloaded function using a set of non-contiguous physical device memory blocks transparently. Further, such techniques may reduce some of the memory garbage collection overhead at the cost of address translations.

(46) In another embodiment, the device manager **112** can perform memory management functions to enforce memory protections. Accordingly, the offloaded functions **118** may need to be protected from accessing each other's memory space. Such privacy and security functionality can allow the disclosed systems to enable secure and robust operation of multiple active offloaded functions. In another embodiment, the device manager **112** may check the addresses of memory transactions of a given offloaded function against the memory address space allocated to it. In one embodiment, the device manager **112** can also provide certain privilege levels so that certain functions can be allowed to access device memory of the other offloaded

functions.

(47) Internal Data Transfer Mechanisms

(48) In some embodiments, data transfer with device memory **116** can occur between the storage controller (not shown) for flash data on flash **114** and the system memory **108**. In other embodiments, device memory **116** can be used by computational program functions that run on the storage device **104**. Since device memory **116** is not exposed/mapped to system memory **108**, the computational storage device **104** can employ the following mechanism to address the internal memory for data transfers described above. In particular, the storage device **104** can employ one or more addressing schemes that is specified to host software (e.g., applications **106**). Further, the host software application interfaces (e.g., drivers **110**) can translate device memory **116** accordingly to instruct the computational storage device **104** on the action to take with the memory **116**. Additionally, the memory **116** can be described for data transfers and for executing a computational program function within the storage device **104**.

(49) Device Memory Addressing Schemes

(50) In some embodiments, the device memory **116** may be described in the following ways (described below), for example, depending on whether the device memory **116** is host **102** managed or storage device **104** managed. As used herein, addressing can refer to a mechanism to describe full or a part of device memory that does not require a mapping into the host's address space, where the mechanism provided is virtualized for local and remote access. With host **102** managed device memory, device memory can be effectively described by a 64-bit address which can address up to about 2.sup.64 or about 18 EB, which is extensible for future usages. Host software (e.g., applications **106**) builds a request to address device memory **116** using this 64-bit address which starts at offset zero that points to the start of memory and then scales through till the end of memory. In one embodiment, the memory increments may be measured at the byte granularity or at a much coarser block granularity which the device may advertise. For example, the disclosed systems can set a block granularity at 512 bytes which means that memory is allocated at the minimal granularity level advertised by the device and also transferred at these same increments to a minimum. As noted, the device memory allocation be host managed or device managed. In case of the device managed device memory, the device provides additional commands to allocate/deallocate memory. In some respects, this mechanism can be similar to block access with disk drives.

(51) FIG. 2 is an illustration of an exemplary schematic of memory allocations with respective offsets in a device memory space, in accordance with example embodiments of the disclosure. Diagram **200** shows three memory allocations **202** (**204**, **206**, and **208**) and their respective 64-bit offsets in the 4GB device memory space. In this host managed mechanism, the device may choose to apply a hash to the address that encodes the address further as a means of protection. Here, the internal device workings may only work with these translated offsets.

(52) FIG. 3 is an illustration of an exemplary schematic describing aspects of device managed device memory, in accordance with example embodiments of the disclosure. In another embodiment, with device managed device memory, the memory that is allocated can be managed by the device itself. As shown in diagram **300**, on any allocation of a required size requested by host software the device can return a token key (e.g., token key **302**) or an opaque memory handle back when successful. In some embodiments, the key along with an offset address may be sufficient to address 2.sup.64 addresses. In some aspects, the memory can include a secure address space, and the host can perform one or more multitenant operations based on the secure address space. In some cases, if the memory is device managed, the address can be secured by the token and/or opaque memory handle where multiple tenants may use the same device memory but with different tokens. For host managed memory, the host can manage multi-tenancy.

(53) As shown in diagram **304**, in various embodiments, the one or more host applications can provide the token key **306** along with the offset within the allocated memory **308** to build the unique device memory address. The token key can be private to the device and can be protected if needed. The offset can range anywhere from 2.sup.12 to 1 less than the minimum bits required for token key. FIG. 3 shows how device memory of 256 KB is allocated and used to create an effective address.

(54) Note on Device Memory

(55) In some aspects, the device may have one large memory available for compute programs. However, the device may also make available, in certain instances, more than one memory type. In this case, the disclosed systems can set aside additional bits to denote the memory type. In some respects, the bits can be encoded in a key token such as the token described above.

(56) Host Interface

(57) In various embodiments, the host software interface library can provide an interface to allocate/deallocate device memory. The interface can include additional options to specify which device memory pool to allocate from. Further, the interface provides an abstracted view, where, the inner details of the implementation are hidden from host user for simplicity. In some embodiments, an application programming interface (API) can take the following definition: csAllocMem(device_instance_handle, memory_flags, allocation_size, &returned_handle). Several of the figures below (e.g., FIGS. 4, 5, 6, and 7) represent diagrams that can be used as part of an API used in connection with the disclosed systems.

(58) FIG. 4 is an illustration of an exemplary data structure that the device may advertise its memory management and usage capability, in accordance with example embodiments of the disclosure.

(59) Discovering New Addressing Capability

(60) In various embodiments, the device may advertise its memory management and usage capability so that host software can utilize it. In one embodiment, NVMe devices may provide these details in a data structure **400** such as an identify controller data structure, since this data structure can represent the first data that is queried by a host device driver. In some embodiments, the device may populate a capability as shown FIG. 4. As shown, the data structure **400** includes a computational storage capabilities field **402** and a device memory data granularity field **404**.

(61) FIG. 5 is an illustration of an exemplary commands for using device memory address for DMA, in accordance with example embodiments of the disclosure. In various embodiments, the disclosed systems can include DMA between device memory and host memory, which can include either a read or a write to device memory, as shown in diagram **500** by the read compute memory **502** and the write compute memory **504** fields. Further, each of these reads or writes can be addressed in the same manner in NVMe for all other fields except the command field.

(62) FIG. 6 is an illustration of an exemplary NVMe command contains transfer details to transfer data between host and device memory which has corresponding fields, in accordance with example embodiments of the disclosure. For example, a protocol-based (e.g., an NVMe-based) Data Pointer Register field (DPTR field) and/or a DMDA field can provide a mechanism to describe host memory. Diagram **600** shows sub-diagram **602** which describes a data pointer (DPTR) field that describes host data pointers, sub-diagram **604** which shows a number of Dwords (NUMD) field that represents the transfer size for command Dwords, and sub-diagram **606** which describes a device memory data address (DMDA) field that represents device data pointers for command Dwords.

(63) In some embodiments, the NVMe command can contain a scatter gather list (SGL) which has the fields as shown and described in FIG. 6. In some aspects, the DPTR field can refer to host memory whose definition is in base specifications. Further, the DMDA field can refer to the device memory specified as a 64-bit address as described above.

(64) Using Device Memory Address for Executing Compute Programs

(65) In various embodiments, compute programs use device memory to access data as input and provide results as output. The device memory may be allocated as described by the schemes described above. Device memory can be represented as 64-bits and the program argument that points can define the memory.

(66) In various embodiments, a compute program as used in connection with the disclosed systems may have one or more arguments which it works with. Arguments may be scalar or vector arguments. Accordingly, the arguments may hold data or be a pointer to device memory. The programs definition has the necessary inputs defined and the command from host will provide that list directly from the host application. The device can interpret each of these arguments as is and pass it down to the program. For example, in the function: foo_function(uint32 size, uint64 src_ptr, unit64

dst_ptr), the host application has passed a 32-bit data value and 2 device memory pointers as part of the command.

(67) FIG. 7 is an illustration of an exemplary special type of scatter gather list (SGL) mechanism is employed to keep memory usage similar for local and remote attached computational storage devices, in accordance with example embodiments of the disclosure. In particular, the device can provide a mechanism for addressing its internal memory used by offloaded programs. The mechanism for some protocols such as Non-Volatile Memory Express (NVMe) can provide an addressing scheme that can be common between a locally attached device (e.g., via Peripheral Component Interconnect Express, PCIe) or a remotely attached device (NVMe over fabric, NVMe-oF). The Host software can make use of this relatively simple and scalable mechanism and need not have to determine how the device is attached while addressing it. In particular, diagram **700** includes an SGL identifier **702** field, a reserved **704** field, a length **706** field, and an offset **708** field. In some aspects, the SGL can provide an SGL identifier **702** to denote the type of descriptor that needs to be used to address device memory. The length **706** field describes the length of the device memory addressed while offset **708** describes the offset within that memory. The reserve **704** field can be kept as a reserved field for future usages.

(68) Using Device Memory Address for Storage I/Os

(69) Regardless of the memory management used to allocate device memory, the disclosed systems describe that memory to the device. In particular, a type of scatter gather list (SGL) mechanism is employed by the disclosed systems to keep memory usage similar for local and remote attached computational storage devices. In various aspects, the system can transfer memory between one device memory location to the next memory location (in the same device). Alternatively or additionally, the system can transfer memory from multiple device locations to the host. In particular, the system can transfer data between one location of device memory to another location in same device memory: certain usages may require transfer of data from one location to the next location. Further, the data may be used by different offloaded programs who may not have access to all memory locations. Moreover, the host in this case can manage the security and multi-tenancy through memory handle abstractions and access restrictions. One example use case can include data loaded from a storage name into device memory. For example, memory A can first get decrypted by the first offloaded program X1 into device memory B. In parallel, another instance of X1 (e.g., an X2 instance) can decrypt device memory C into device memory D. The host may then copy memory B and memory D in device memory location E and present it to offload program Y, for example, to analyze and filter memory B and memory D. In other aspects, the system can transfer between two separate device memory locations: here, the device may have more than one device memories. Additionally, the device can transfer memory between multiple scattered device memory locations to one device memory location on same device memory, which can represent an SGL usage. Further, the device can transfer memory between multiple scattered device memory locations to host memory.

(70) In various embodiments, a SGL Descriptor Type set to a predetermined value (e.g., 9) can denote a type for Compute Memory for Data SGL descriptor. Further, an SGL Descriptor Sub Type for all transfers can be set to another value (e.g., 1) and can operate using the Offset field as an offset from the start of Compute Memory for Data. For example, the Offset field contains an offset from the beginning of the location where Compute Memory for Data is located and whose length defined in Length is governed by the block granularity field. The Offset field which describes a 64-bit address shall be used as described for memory managed by host or device. In another embodiment, various commands (e.g., NVMe I/O commands) may use the SGLs to describe Compute Memory for Data for their operation.

(71) FIG. 8 is an illustration of an exemplary table **800** that summarizes interaction of data buffers with NVMe I/O commands, in accordance with example embodiments of the disclosure. The NVMe read and write commands **802** can be defined with corresponding physical region page (PRP) and SGL usages. In particular, embodiments of this disclosure describe load and store commands to transfer data between storage namespaces and device memory. Alternately, NVMe read and write commands may be modified to include using the disclosed device memory addressing scheme. The updated SGL can be substituted for those cases where device memory is addressed.

(72) FIG. 9 is an illustration of an exemplary SGL descriptor for data transfer, in accordance with example embodiments of the disclosure. The following example demonstrates the usage of the described SGL descriptor for data transfer. Diagram **900** shows an SQE **902**, the compute memory for data SGL **904**, and a device memory for compute **906**. The compute memory for data SGL **904** includes predetermined offset **908**, length **910**, and SGL identifier **912** fields. Further, the device memory for compute **906** includes a 64 KB memory **914**. In some cases, NVMe commands can be transferred from the host to device using descriptors defined for each command. These descriptors can be populated in an NVMe submission queue (SQ). Each SQ may support multiple submission queue entries (SQEs). Host software can transfer each request as a descriptor to a free SQE and lets the device know about the work posted to the SQ by ringing its doorbell. In particular, diagram **900** shows that the SQE **902** entry can be populated with a standard NVMe read command and that the Data Pointer (DPTR field) can be updated with the SGL entry that points to device memory (e.g., the 4 KB memory **914**). Here, the SGL defines a device memory block at a predetermined offset **908** and a length **910** of 64 KB. The other fields of the SQE are populated with a regular NVMe read. An NVMe read using this SGL can populate the storage read contents directly at the predetermined offset **908**. The storage controller can interpret this inner definition.

(73) The mechanisms described herein with this disclosure provide a unique methods to address and access device memory for computational storage devices. These mechanisms are adaptable for local and remote attached devices and are scalable.

(74) FIG. 10 is an illustration of an exemplary operational flow **1000** illustrating example operations associated with the disclosed systems, in accordance with example embodiments of the disclosure. In particular, the operational flow **1000** is from the point of view of the system (e.g., as opposed to a host device perspective and/or a storage device perspective). At block **1002**, the disclosed systems can establish a connection via an interface (e.g., an API), between a host device and a storage device. At block **1004**, the disclosed systems can transfer data, via the interface, between first memory associated with the host device and second memory associated with the storage device by performing a data operation on the second memory by an application executed by the host. Further, the storage device can include a processing element that accelerates the data operation by performing at least one offload function on the data operation.

(75) FIG. 11 is an illustration of another exemplary operational flow **1100** illustrating example operations associated with the disclosed systems, in accordance with example embodiments of the disclosure. In particular, the operational flow **1100** is from the point of view of the storage device (e.g., as opposed to a host device perspective and/or a general system perspective). In particular, at block **1102**, the disclosed systems can transmit information to the host device to establish a connection via an interface between the storage device and a host device. At block **1104**, the disclosed systems can receive a data request from the host device. At block **1106**, the disclosed systems can perform a data operation, based on the data request, on the storage device memory by performing at least one offload function on the data operation. At block **1108**, the disclosed systems can provide data resulting from the data operation, via the interface, to the host device.

(76) FIG. 12 shows an example schematic diagram of a system that can be used to practice embodiments of the present disclosure. As shown in FIG. 12, this particular embodiment may include one or more management computing entities **1200**, one or more networks **1205**, and one or more user devices **1210** (e.g., host devices, storage devices, additional devices, etc.). In various embodiments, the management computing entities **1200** can be configured to run operations associated with the disclosed systems and associated embodiments described throughout the disclosure. Further, the management computing entities **1200** can reside in any suitable portion of the disclosed systems (e.g., such as a storage device manager, host application, a remote host, another device, combinations thereof, and/or the like). Each of these components, entities, devices, systems, and similar words used herein interchangeably may be in direct or indirect communication with, for example, one another over the same or different wired or wireless networks. Additionally, while FIG. 12 illustrates the various system entities as separate, standalone entities, the various embodiments are not limited to this particular architecture. As noted, the communications can be performed using the any suitable protocols described further herein.

(77) FIG. 13 shows an example schematic diagram **1300** of a management computing entity, in accordance with example embodiments of the disclosure. As noted, the management computing entity can be configured to run operations associated with the disclosed systems and associated embodiments described throughout the disclosure. Further, the management computing entities **1200** can reside in any suitable portion of the

disclosed systems. In particular, a content component may serve to determine signals indicative of data (e.g., video, audio, text, data, combinations thereof, and/or the like) to be transmitted over the system described herein. In another embodiment, the determination of the signal for transmission may be, for example, based on a user input to a device, a predetermined schedule of data transmissions on a network associated with the system, changes in network conditions, and the like. In one embodiment, the signal may include that data may be encapsulated in a data frame and/or packet that is configured to be sent from a device to one or more devices on the network.

(78) In another embodiment, the processing element **1305** may serve to determine various parameters associated data transmitted over the network associated with the disclosed systems. As another example, the processing element **1305** may serve perform various acceleration operations such as at least portions of an offload functionality, data pre- or post-processing, combinations thereof, and/or the like.

(79) In one embodiment, a transmitting component (not shown) may serve to transmit the signal from one device to another device on the network. For example, the transmitting component may serve to prepare a transmitter (e.g., transmitter **1204** of FIG. **12**, below) to transmit the signal over the network. For example, the transmitting component may queue data in one or more buffers, may ascertain that the transmitting device and associated transmitters are functional and have adequate power to transmit the signal over the network, may adjust one or more parameters (e.g., modulation type, signal amplification, signal power level, noise rejection, combinations thereof, and/or the like) associated with the transmission of the data.

(80) In general, the terms computing entity, computer, entity, device, system, and/or similar words used herein interchangeably may refer to, for example, one or more computers, computing entities, desktop computers, mobile phones, tablets, phablets, notebooks, laptops, distributed systems, gaming consoles (for example Xbox, Play Station, Wii), watches, glasses, iBeacons, proximity beacons, key fobs, radio frequency identification (RFID) tags, ear pieces, scanners, televisions, dongles, cameras, wristbands, wearable items/devices, kiosks, input terminals, servers or server networks, blades, gateways, switches, processing devices, processing entities, set-top boxes, relays, routers, network access points, base stations, the like, and/or any combination of devices or entities adapted to perform the functions, operations, and/or processes described herein. Such functions, operations, and/or processes may include, for example, transmitting, receiving, operating on, processing, displaying, storing, determining, creating/generating, monitoring, evaluating, comparing, and/or similar terms used herein interchangeably. In one embodiment, these functions, operations, and/or processes can be performed on data, content, information, and/or similar terms used herein interchangeably.

(81) As indicated, in one embodiment, the management computing entity **1200** may also include one or more communications interfaces **1320** for communicating with various computing entities, such as by communicating data, content, information, and/or similar terms used herein interchangeably that can be transmitted, received, operated on, processed, displayed, stored, and/or the like. For instance, the management computing entity **1200** may communicate with user devices **1210** and/or a variety of other computing entities.

(82) As shown in FIG. **13**, in one embodiment, the management computing entity **1200** may include or be in communication with one or more processing elements **1305** (also referred to as processors, processing circuitry, and/or similar terms used herein interchangeably) that communicate with other elements within the management computing entity **1200** via a bus, for example. As will be understood, the processing element **1305** may be embodied in a number of different ways. For example, the processing element **1305** may be embodied as one or more complex programmable logic devices (CPLDs), microprocessors, multi-core processors, coprocessing entities, application-specific instruction-set processors (ASIPs), microcontrollers, and/or controllers. Further, the processing element **1305** may be embodied as one or more other processing devices or circuitry. The term circuitry may refer to an entirely hardware embodiment or a combination of hardware and computer program products. Thus, the processing element **1305** may be embodied as integrated circuits, application specific integrated circuits (ASICs), field programmable gate arrays (FPGAs), programmable logic arrays (PLAs), hardware accelerators, other circuitry, and/or the like. As will therefore be understood, the processing element **1305** may be configured for a particular use or configured to execute instructions stored in volatile or non-volatile media or otherwise accessible to the processing element **1305**. As such, whether configured by hardware or computer program products, or by a combination thereof, the processing element **1305** may be capable of performing steps or operations according to embodiments of the present disclosure when configured accordingly.

(83) In one embodiment, the management computing entity **1200** may further include or be in communication with non-volatile media (also referred to as non-volatile storage, memory, memory storage, memory circuitry and/or similar terms used herein interchangeably). In one embodiment, the non-volatile storage or memory may include one or more non-volatile storage or memory media **1310**, including but not limited to hard disks, ROM, PROM, EPROM, EEPROM, flash memory, MMCs, SD memory cards, Memory Sticks, CBRAM, PRAM, FeRAM, NVRAM, MRAM, RRAM, SONOS, FJG RAM, Millipede memory, racetrack memory, and/or the like. As will be recognized, the non-volatile storage or memory media may store databases, database instances, database management systems, data, applications, programs, program components, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like. The term database, database instance, database management system, and/or similar terms used herein interchangeably may refer to a collection of records or data that is stored in a computer-readable storage medium using one or more database models, such as a hierarchical database model, network model, relational model, entity—relationship model, object model, document model, semantic model, graph model, and/or the like.

(84) In one embodiment, the management computing entity **1200** may further include or be in communication with volatile media (also referred to as volatile storage, memory, memory storage, memory circuitry and/or similar terms used herein interchangeably). In one embodiment, the volatile storage or memory may also include one or more volatile storage or memory media **1315**, including but not limited to RAM, DRAM, SRAM, FPM DRAM, EDO DRAM, SDRAM, DDR SDRAM, DDR2 SDRAM, DDR3 SDRAM, RDRAM, TTRAM, T-RAM, Z-RAM, RIMM, DIMM, SIMM, VRAM, cache memory, register memory, and/or the like. As will be recognized, the volatile storage or memory media may be used to store at least portions of the databases, database instances, database management systems, data, applications, programs, program components, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like being executed by, for example, the processing element **1305**. Thus, the databases, database instances, database management systems, data, applications, programs, program components, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like may be used to control certain aspects of the operation of the management computing entity **1200** with the assistance of the processing element **1305** and operating system.

(85) As indicated, in one embodiment, the management computing entity **1200** may also include one or more communications interfaces **1320** for communicating with various computing entities, such as by communicating data, content, information, and/or similar terms used herein interchangeably that can be transmitted, received, operated on, processed, displayed, stored, and/or the like. Such communication may be executed using a wired data transmission protocol, such as peripheral component interconnect express (PCIe), fiber distributed data interface (FDDI), digital subscriber line (DSL), Ethernet, asynchronous transfer mode (ATM), frame relay, data over cable service interface specification (DOCSIS), or any other wired transmission protocol. Similarly, the management computing entity **1200** may be configured to communicate via wireless external communication networks using any of a variety of protocols, such as general packet radio service (GPRS), Universal Mobile Telecommunications System (UMTS), Code Division Multiple Access 2000 (CDMA2000), CDMA2000 1× (1×RTT), Wideband Code Division Multiple Access (WCDMA), Time Division-Synchronous Code Division Multiple Access (TD-SCDMA), Long Term Evolution (LTE), Evolved Universal Terrestrial Radio Access Network (E-UTRAN), Evolution-Data Optimized (EVDO), High Speed Packet Access (HSPA), High-Speed Downlink Packet Access (HSDPA), IEEE 802.11 (Wi-Fi), Wi-Fi Direct, 802.16 (WiMAX), ultra-wideband (UWB), infrared (IR) protocols, near field communication (NFC) protocols, ZigBee, Bluetooth protocols, 5G protocol, wireless universal serial bus (USB) protocols, and/or any other wireless protocol.

(86) Although not shown, the management computing entity **1200** may include or be in communication with one or more input elements, such as a keyboard input, a mouse input, a touch screen/display input, motion input, movement input, audio input, pointing device input, joystick input, keypad input, and/or the like. The management computing entity **1200** may also include or be in communication with one or more output elements (not shown), such as audio output, video output, screen/display output, motion output, movement output, and/or the like.

(87) As will be appreciated, one or more of the management computing entity's **1200** components may be located separately from other management computing entity **1200** components, such as in a distributed system. Furthermore, one or more of the components may be combined and additional components performing functions described herein may be included in the management computing entity **1200**. Thus, the management computing entity **1200** can be adapted to accommodate a variety of needs and circumstances. As will be recognized, these architectures and descriptions are provided for example purposes only and are not limiting to the various embodiments.

(88) A user may be an individual, a family, a company, an organization, an entity, a department within an organization, a representative of an organization and/or person, and/or the like. In one example, users may be employees, residents, customers, and/or the like. For instance, a user may operate a user device **1210** that includes one or more components that are functionally similar to those of the management computing entity **1200**.

(89) In various aspects, the processing component, the transmitting component, and/or the receiving component (not shown) may be configured to operate on one or more may include aspects of the functionality of the management computing entity **1200**, as shown and described in connection with FIGS. **12** and **13** here. In particular, the processing component, the transmitting component, and/or the receiving component may be configured to be in communication with one or more processing elements **1305**, memory **1310**, volatile memory **1315**, and may include a communication interface **1320** (e.g., to facilitate communication between devices).

(90) FIG. **14** shows an example schematic diagram of a user device, in accordance with example embodiments of the disclosure. FIG. **14** provides an illustrative schematic representative of a user device **1410** (e.g., a host device, a storage device, a peripheral device, etc.) that can be used in conjunction with embodiments of the present disclosure. In general, the terms device, system, computing entity, entity, and/or similar words used herein interchangeably may refer to, for example, one or more computers, computing entities, desktops, mobile phones, tablets, phablets, notebooks, laptops, distributed systems, gaming consoles (for example Xbox, Play Station, Wii), watches, glasses, key fobs, radio frequency identification (RFID) tags, ear pieces, scanners, cameras, wristbands, kiosks, input terminals, servers or server networks, blades, gateways, switches, processing devices, processing entities, set-top boxes, relays, routers, network access points, base stations, the like, and/or any combination of devices or entities adapted to perform the functions, operations, and/or processes described herein. User devices **1410** can be operated by various parties. As shown in FIG. **14**, the user device **1410** can include an antenna **1412**, a transmitter **1404** (for example radio), a receiver **1406** (for example radio), and a processing element **1408** (for example CPLDs, FPGAs, microprocessors, multi-core processors, coprocessing entities, ASIPs, microcontrollers, and/or controllers) that provides signals to and receives signals from the transmitter **1404** and receiver **1406**, respectively.

(91) The signals provided to and received from the transmitter **1404** and the receiver **1406**, respectively, may include signaling information in accordance with air interface standards of applicable wireless systems. In this regard, the user device **1410** may be capable of operating with one or more air interface standards, communication protocols, modulation types, and access types. More particularly, the user device **1410** may operate in accordance with any of a number of wireless communication standards and protocols, such as those described above with regard to the management computing entity **1400** of FIG. **14**. In a particular embodiment, the user device **1410** may operate in accordance with multiple wireless communication standards and protocols, such as the disclosed IoT DOCSIS protocol, UMTS, CDMA2000, 1×RTT, WCDMA, TD-SCDMA, LTE, E-UTRAN, EVDO, HSPA, HSDPA, 5G, Wi-Fi, Wi-Fi Direct, WiMAX, UWB, IR, NFC, Bluetooth, USB, and/or the like. Similarly, the user device **1410** may operate in accordance with multiple wired communication standards and protocols, such as those described above with regard to the management computing entity **1400** via a network interface **1420**.

(92) Via these communication standards and protocols, the user device **1410** can communicate with various other entities using concepts such as Unstructured Supplementary Service Data (USSD), Short Message Service (SMS), Multimedia Messaging Service (MMS), Dual-Tone Multi-Frequency Signaling (DTMF), and/or Subscriber Identity Component Dialer (SIM dialer). The user device **1410** can also download changes, additions, and updates, for instance, to its firmware, software (for example including executable instructions, applications, program components), and operating system.

(93) According to one embodiment, the user device **1410** may include location determining aspects, devices, components, functionalities, and/or similar words used herein interchangeably. The location determining aspects may be used to inform the models used by the management computing entity and one or more of the models and/or machine learning techniques described herein. For example, the user device **1410** may include outdoor positioning aspects, such as a location component adapted to acquire, for example, latitude, longitude, altitude, geocode, course, direction, heading, speed, universal time (UTC), date, and/or various other information/data. In one embodiment, the location component can acquire data, sometimes known as ephemeris data, by identifying the number of satellites in view and the relative positions of those satellites. The satellites may be a variety of different satellites, including Low Earth Orbit (LEO) satellite systems, Department of Defense (DOD) satellite systems, the European Union Galileo positioning systems, the Chinese Compass navigation systems, Indian Regional Navigational satellite systems, and/or the like. Alternatively, the location information can be determined by triangulating the user device's **1410** position in connection with a variety of other systems, including cellular towers, Wi-Fi access points, and/or the like. Similarly, the user device **1410** may include indoor positioning aspects, such as a location component adapted to acquire, for example, latitude, longitude, altitude, geocode, course, direction, heading, speed, time, date, and/or various other information/data. Some of the indoor systems may use various position or location technologies including RFID tags, indoor beacons or transmitters, Wi-Fi access points, cellular towers, nearby computing devices (for example smartphones, laptops) and/or the like. For instance, such technologies may include the iBeacons, Gimbal proximity beacons, Bluetooth Low Energy (BLE) transmitters, NFC transmitters, and/or the like. These indoor positioning aspects can be used in a variety of settings to determine the location of someone or something to within inches or centimeters.

(94) The user device **1410** may also comprise a user interface (that can include a display **1416** coupled to a processing element **1408**) and/or a user input interface (coupled to a processing element **1408**). For example, the user interface may be a user application, browser, user interface, and/or similar words used herein interchangeably executing on and/or accessible via the user device **1410** to interact with and/or cause display of information from the management computing entity **1400**, as described herein. The user input interface can comprise any of a number of devices or interfaces allowing the user device **1410** to receive data, such as a keypad **1418** (hard or soft), a touch display, voice/speech or motion interfaces, or other input devices. In embodiments including a keypad **1418**, the keypad **1418** can include (or cause display of) the conventional numeric (0-9) and related keys (#, *), and other keys used for operating the user device **1410** and may include a full set of alphabetic keys or set of keys that may be activated to provide a full set of alphanumeric keys. In addition to providing input, the user input interface can be used, for example, to activate or deactivate certain functions, such as screen savers and/or sleep modes.

(95) The user device **1410** can also include volatile storage or memory **1422** and/or non-volatile storage or memory **1424**, which can be embedded and/or may be removable. For example, the non-volatile memory may be ROM, PROM, EPROM, EEPROM, flash memory, MMCs, SD memory cards, Memory Sticks, CBRAM, PRAM, FeRAM, NVRAM, MRAM, RRAM, SONOS, FJG RAM, Millipede memory, racetrack memory, and/or the like. The volatile memory may be RAM, DRAM, SRAM, FPM DRAM, EDO DRAM, SDRAM, DDR SDRAM, DDR2 SDRAM, DDR3 SDRAM, RDRAM, TTRAM, T-RAM, Z-RAM, RIMM, DIMM, SIMM, VRAM, cache memory, register memory, and/or the like. The volatile and non-volatile storage or memory can store databases, database instances, database management systems, data, applications, programs, program components, scripts, source code, object code, byte code, compiled code, interpreted code, machine code, executable instructions, and/or the like to implement the functions of the user device **1410**. As indicated, this may include a user application that is resident on the entity or accessible through a browser or other user interface for communicating with the management computing entity **1400** and/or various other computing entities.

(96) In another embodiment, the user device **1410** may include one or more components or functionality that are the same or similar to those of the management computing entity **1400**, as described in greater detail above. As will be recognized, these architectures and descriptions are provided for example purposes only and are not limiting to the various embodiments.

(97) Certain embodiments may be implemented in one or a combination of hardware, firmware, and software. Other embodiments may also be

implemented as instructions stored on a computer-readable storage device, which may be read and executed by at least one processor to perform the operations described herein. A computer-readable storage device may include any non-transitory memory mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a computer-readable storage device may include read-only memory (ROM), random-access memory (RAM), magnetic disk storage media, optical storage media, flash-memory devices, and other storage devices and media.

(98) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. The terms “computing device”, “user device”, “communication station”, “station”, “handheld device”, “mobile device”, “wireless device” and “user equipment” (UE) as used herein refers to a wireless communication device such as a cellular telephone, smartphone, tablet, netbook, wireless terminal, laptop computer, a femtocell, High Data Rate (HDR) subscriber station, access point, printer, point of sale device, access terminal, or other personal communication system (PCS) device. The device may be either mobile or stationary.

(99) As used within this document, the term “communicate” is intended to include transmitting, or receiving, or both transmitting and receiving. This may be particularly useful in claims when describing the organization of data that is being transmitted by one device and received by another, but only the functionality of one of those devices is required to infringe the claim. Similarly, the bidirectional exchange of data between two devices (both devices transmit and receive during the exchange) may be described as ‘communicating’, when only the functionality of one of those devices is being claimed. The term “communicating” as used herein with respect to a wireless communication signal includes transmitting the wireless communication signal and/or receiving the wireless communication signal. For example, a wireless communication unit, which is capable of communicating a wireless communication signal, may include a wireless transmitter to transmit the wireless communication signal to at least one other wireless communication unit, and/or a wireless communication receiver to receive the wireless communication signal from at least one other wireless communication unit.

(100) Some embodiments may be used in conjunction with various devices and systems, for example, a Personal Computer (PC), a desktop computer, a mobile computer, a laptop computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, a Personal Digital Assistant (PDA) device, a handheld PDA device, an on-board device, an off-board device, a hybrid device, a vehicular device, a non-vehicular device, a mobile or portable device, a consumer device, a non-mobile or non-portable device, a wireless communication station, a wireless communication device, a wireless Access Point (AP), a wired or wireless router, a wired or wireless modem, a video device, an audio device, an audio-video (A/V) device, a wired or wireless network, a wireless area network, a Wireless Video Area Network (WVAN), a Local Area Network (LAN), a Wireless LAN (WLAN), a Personal Area Network (PAN), a Wireless PAN (WPAN), and the like.

(101) Some embodiments may be used in conjunction with one way and/or two-way radio communication systems, cellular radio-telephone communication systems, a mobile phone, a cellular telephone, a wireless telephone, a Personal Communication Systems (PCS) device, a PDA device which incorporates a wireless communication device, a mobile or portable Global Positioning System (GPS) device, a device which incorporates a GPS receiver or transceiver or chip, a device which incorporates an RFID element or chip, a Multiple Input Multiple Output (MIMO) transceiver or device, a Single Input Multiple Output (SIMO) transceiver or device, a Multiple Input Single Output (MISO) transceiver or device, a device having one or more internal antennas and/or external antennas, Digital Video Broadcast (DVB) devices or systems, multi-standard radio devices or systems, a wired or wireless handheld device, e.g., a Smartphone, a Wireless Application Protocol (WAP) device, or the like.

(102) Some embodiments may be used in conjunction with one or more types of wireless communication signals and/or systems following one or more wireless communication protocols, for example, Radio Frequency (RF), Infrared (IR), Frequency-Division Multiplexing (FDM), Orthogonal FDM (OFDM), Time-Division Multiplexing (TDM), Time-Division Multiple Access (TDMA), Extended TDMA (E-TDMA), General Packet Radio Service (GPRS), extended GPRS, Code-Division Multiple Access (CDMA), Wideband CDMA (WCDMA), CDMA 2000, single-carrier CDMA, multi-carrier CDMA, Multi-Carrier Modulation (MDM), Discrete Multi-Tone (DMT), Bluetooth™, Global Positioning System (GPS), Wi-Fi, Wi-Max, ZigBee™, Ultra-Wideband (UWB), Global System for Mobile communication (GSM), 2G, 2.5G, 3G, 3.5G, 4G, Fifth Generation (5G) mobile networks, 3GPP, Long Term Evolution (LTE), LTE advanced, Enhanced Data rates for GSM Evolution (EDGE), or the like. Other embodiments may be used in various other devices, systems, and/or networks.

(103) Although an example processing system has been described above, embodiments of the subject matter and the functional operations described herein can be implemented in other types of digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them.

(104) Embodiments of the subject matter and the operations described herein can be implemented in digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Embodiments of the subject matter described herein can be implemented as one or more computer programs, i.e., one or more components of computer program instructions, encoded on computer storage medium for execution by, or to control the operation of, information/data processing apparatus. Alternatively, or in addition, the program instructions can be encoded on an artificially-generated propagated signal, for example a machine-generated electrical, optical, or electromagnetic signal, which is generated to encode information/data for transmission to suitable receiver apparatus for execution by an information/data processing apparatus. A computer storage medium can be, or be included in, a computer-readable storage device, a computer-readable storage substrate, a random or serial access memory array or device, or a combination of one or more of them. Moreover, while a computer storage medium is not a propagated signal, a computer storage medium can be a source or destination of computer program instructions encoded in an artificially-generated propagated signal. The computer storage medium can also be, or be included in, one or more separate physical components or media (for example multiple CDs, disks, or other storage devices).

(105) The operations described herein can be implemented as operations performed by an information/data processing apparatus on information/data stored on one or more computer-readable storage devices or received from other sources.

(106) The term “data processing apparatus” encompasses all kinds of apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, a system on a chip, or multiple ones, or combinations, of the foregoing. The apparatus can include special purpose logic circuitry, for example an FPGA (field programmable gate array) or an ASIC (application-specific integrated circuit). The apparatus can also include, in addition to hardware, code that creates an execution environment for the computer program in question, for example code that constitutes processor firmware, a protocol stack, a database management system, an operating system, a cross-platform runtime environment, a virtual machine, or a combination of one or more of them. The apparatus and execution environment can realize various different computing model infrastructures, such as web services, distributed computing and grid computing infrastructures.

(107) A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, declarative or procedural languages, and it can be deployed in any form, including as a stand-alone program or as a component, component, subroutine, object, or other unit suitable for use in a computing environment. A computer program may, but need not, correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or information/data (for example one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (for example files that store one or more components, sub-programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

(108) The processes and logic flows described herein can be performed by one or more programmable processors executing one or more computer programs to perform actions by operating on input information/data and generating output. Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital

computer. Generally, a processor will receive instructions and information/data from a read-only memory or a random access memory or both. The essential elements of a computer are a processor for performing actions in accordance with instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive information/data from or transfer information/data to, or both, one or more mass storage devices for storing data, for example magnetic, magneto-optical disks, or optical disks. However, a computer need not have such devices. Devices suitable for storing computer program instructions and information/data include all forms of non-volatile memory, media and memory devices, including by way of example semiconductor memory devices, for example EPROM, EEPROM, and flash memory devices; magnetic disks, for example internal hard disks or removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

(109) To provide for interaction with a user, embodiments of the subject matter described herein can be implemented on a computer having a display device, for example a CRT (cathode ray tube) or LCD (liquid crystal display) monitor, for displaying information/data to the user and a keyboard and a pointing device, for example a mouse or a trackball, by which the user can provide input to the computer. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, for example visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser.

(110) Embodiments of the subject matter described herein can be implemented in a computing system that includes a back-end component, for example as an information/data server, or that includes a middleware component, for example an application server, or that includes a front-end component, for example a client computer having a graphical user interface or a web browser through which a user can interact with an embodiment of the subject matter described herein, or any combination of one or more such back-end, middleware, or front-end components. The components of the system can be interconnected by any form or medium of digital information/data communication, for example a communication network. Examples of communication networks include a local area network ("LAN") and a wide area network ("WAN"), an inter-network (for example the Internet), and peer-to-peer networks (for example ad hoc peer-to-peer networks).

(111) The computing system can include clients and servers. A client and server are generally remote from each other and typically interact through a communication network. The relationship of client and server arises by virtue of computer programs running on the respective computers and having a client-server relationship to each other. In some embodiments, a server transmits information/data (for example an HTML page) to a client device (for example for purposes of displaying information/data to and receiving user input from a user interacting with the client device). Information/data generated at the client device (for example a result of the user interaction) can be received from the client device at the server.

(112) While this specification contains many specific embodiment details, these should not be construed as limitations on the scope of any embodiment or of what may be claimed, but rather as descriptions of features specific to particular embodiments. Certain features that are described herein in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

(113) Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. In certain circumstances, multitasking and parallel processing may be advantageous. Moreover, the separation of various system components in the embodiments described above should not be understood as requiring such separation in all embodiments, and it should be understood that the described program components and systems can generally be integrated together in a single software product or packaged into multiple software products.

(114) Thus, particular embodiments of the subject matter have been described. Other embodiments are within the scope of the following claims. In some cases, the actions recited in the claims can be performed in a different order and still achieve desirable results. In addition, the processes depicted in the accompanying figures do not necessarily require the particular order shown, or sequential order, to achieve desirable results. In certain embodiments, multitasking and parallel processing may be advantageous.

(115) Many modifications and other embodiments of the disclosure set forth herein will come to mind to one skilled in the art to which these embodiments pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the embodiments are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A method to manage memory, the method comprising: establishing a connection, via an interface, between a host device and a storage device; and transferring data, via the interface, from first memory associated with the host device to second memory associated with the storage device, wherein the interface comprises an application programming interface (API), and the host device allocates the second memory using at least one command associated with the API; wherein the storage device includes a flash storage and a processing element that accelerates a data operation by performing at least one offload function on the data operation of an application, wherein the data operation is performed on the data by the storage device, wherein the data is transferred between the first memory and the second memory based at least in part on the at least one offload function, and wherein the at least one offload function is based at least in part on the data operation of the application.
2. The method of claim 1, wherein the host device allocates the second memory using a fine-grained allocation scheme or a coarse-grained allocation scheme.
3. The method of claim 1, wherein the method further comprises addressing the second memory and the transferring data is based at least in part on the addressing of the second memory.
4. The method of claim 1, wherein the second memory comprises a secure address space, wherein the method further comprises host device performing one or more multitenant operations based on the secure address space, and wherein the secure address space protects the data of a first tenant from access by a second tenant.
5. The method of claim 1, wherein the storage device comprises a non-volatile memory express (NVMe) enabled storage device and the data operation is performed using a scatter gather list (SGL).
6. The method of claim 1, wherein the performing the data operation further comprises the storage device transferring data from persistent memory associated with the storage device to the second memory.
7. The method of claim 1, wherein the transferring data further comprises the host device providing information regarding the first memory's layout to the storage device.
8. The method of claim 1, wherein the transferring data further comprises the storage device providing information regarding the storage device's memory layout for read and write operations to the host device.
9. The method of claim 1, wherein the establishing the connection comprises a discovery and setup process.

10. The method of claim 1, wherein the offload function accesses a result of the data operation from the second memory associated with the storage device.

11. A non-transitory computer-readable medium storing computer-executable instructions for managing memory which, when executed by a processor, cause the processor to perform operations comprising: establishing a connection, via an interface, between a host device and a storage device; and transferring data, via the interface, from first memory associated with the host device to second memory associated with the storage device, wherein the interface comprises an application programming interface (API), and the host device allocates the second memory using at least one command associated with the API; wherein the storage device includes a flash storage and a processing element that accelerates a data operation by performing at least one offload function on the data operation of an application, wherein the data operation is performed on the data by the storage device, wherein the data is transferred between the first memory and the second memory based at least in part on the at least one offload function, and wherein the at least one offload function is based at least in part on the data operation of the application.

12. The non-transitory computer-readable medium of claim 11, wherein the computer-executable instructions further cause the processor to perform operations comprising addressing the second memory, and the transferring data is based at least in part on the addressing of the second memory.

13. The non-transitory computer-readable medium of claim 11, wherein the second memory comprises a secure address space, wherein the host device performs one or more multitenant operations based on the secure address space and wherein the secure address space protects the data of a first tenant from access by a second tenant.

14. The non-transitory computer-readable medium of claim 11, wherein the storage device comprises a non-volatile memory express (NVMe) enabled storage device and the data operation is performed using a scatter gather list (SGL).

15. The non-transitory computer-readable medium of claim 11, wherein the data operation further comprises the storage device transferring data from persistent memory associated with the storage device to the second memory.

16. A system for managing memory, comprising: a host device comprising a first memory and an application; a storage device comprising a device manager, a flash storage, a processing element, and a second memory; wherein the system stores computer-executable instructions which, when executed by a processor, cause the processor to perform operations comprising: establishing a connection, via an interface, between the host device and the storage device; and transferring data, via the interface, from first memory associated with the host device to second memory associated with the storage device, wherein the interface comprises an application programming interface (API), and the host device allocates the second memory using at least one command associated with the API; wherein the device manager configures the processing element to accelerate a data operation by performing at least one offload function on the data operation of an application, wherein the data operation is performed on the data by the processing element, wherein the data is transferred between the first memory and the second memory based at least in part on the at least one offload function, and wherein the at least one offload function is based at least in part on the data operation of the application.

17. The system of claim 16, wherein the storage device transfers second data, via the interface, between at least a third memory associated with the storage device and a fourth memory associated with the storage device.

18. The system of claim 16, wherein the storage device transfers second data, via the interface, between at least a third memory associated with the storage device and a fourth memory associated with the host device.
